ROBOARM — FULL WIRING (REV D)

Every board pin shown. Wires drawn between parts. Power rails use KiCad power symbols (+24V, +5V, +6V, GND). Unused pins carry no-connect flags.

UART bus A: Mega Serial2 (TX2 D16 via 1k, RX2 D17) -> PDN/UART of DRV1–3 (addr 0…2).
UART bus B: Mega Serial3 (TX3 D14 via 1k, RX3 D15) -> DRV4–5. Gives per-joint load (StallGuard), temperature and open-wire flags.

E-STOP: pressing SW1 opens the relay coil circuit, K1 drops out
and the red 24 V bus goes dead. PI and Mega stay on.
A broken E-stop wire also stops the motors (fail-safe).

Joint encoders: AS5600 reads a magnet on each joint OUTPUT shaft (after the gearbox), so the angle is absolute at power-up.
OUT = analog angle (0–5 V) to Mega A0–A4 through 1k + 100 nF at the Mega end. DIR tied to GND sets direction.
Check the module runs at 5 V (VCC) before buying; SDA/SCL left free for later I2C use.

PS1
Lab bench supply: 24 V, 3 A limit (later: Mean Well LRS-200-24)

F0
10 A main

K1
24 V 30 A relay

SW1
E-STOP (NC, latching)

D1
1N4007

+24V

GND

PS1 for first tests: existing bench supply, 24 V, current limit 3 A.
Its output on/off button cuts motor power only; PI and Mega stay on.
The 24 V goes ONLY to the motor drivers and the servo buck. The PI has its own USB-C supply (PS2).

24 V BUS (red): bench supply + -> F0 -> VM on every driver. Ground return (black): every driver GND -> bench supply −

PS2
Official PI 27 W USB-C supply

U1
Raspberry Pi 5 8GB

PI USB-A to Mega USB-B cable:
powers the Mega and carries commands.

U2
Arduino Mega 2560 R3

R31
1k (UART-TX)

R21
10k

DRV1
TMC2209 module (StepStick)

C20
100uF 35V

M1
17HS19-1684S-PG5
J1 Base: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS

R22
10k

DRV2
TMC2209 module (StepStick)

C21
100uF 35V

M2
17HS19-1684S-PG14
J2 Shoulder: 0.52 Nm 1.68 A × 13.73:1, set 1.4 A RMS

R23
10k

DRV3
TMC2209 module (StepStick)

C22
100uF 35V

M3
17HS19-1684S-PG5
J3 Elbow: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS

R32
1k (UART-TX)

R24
10k

DRV4
TMC2209 module (StepStick)

C23
100uF 35V

M4
17HS08-1004S
J4 Wrist: 0.16 Nm 1 A, set 1.0 A RMS

R25
10k

DRV5
TMC2209 module (StepStick)

C24
100uF 35V

M5
11HS20-0674S
J5 Wrist twist: 0.12 Nm 0.67 A, set 0.67 A RMS

R1
1k
C1
100nF
ENC1
J1 encoder, AS5600 module

R2
1k
C2
100nF
ENC2
J2 encoder, AS5600 module

R3
1k
C3
100nF
ENC3
J3 encoder, AS5600 module

R4
1k
C4
100nF
ENC4
J4 encoder, AS5600 module

R5
1k
C5
100nF
ENC5
J5 encoder, AS5600 module

FAN1
40 mm 5 V fan over driver row

F1
2 A
U6
XL4015 buck, set 6.0 V
F2
3 A
C7
1000uF
C8
100nF
R13
1k
J6
Tool plug, GX12 6-pin
PWM
MG996R gripper servo

Gripper power: 24 V bus -> F1 -> U6 buck (set 6.0 V with a meter BEFORE connecting the servo) -> F2 -> 6 V (orange) -> C7, C8, servo.

BILL OF MATERIALS (electrical parts; full list, links and prices in hardware/bom)

Ref	Qty	Part	Note
U1	1	Raspberry Pi 5 8GB	owned
PS2	1	Official Raspberry Pi 27 W USB-C supply	owned
U2	1	Arduino Mega 2560 R3	owned
PS1	1	24 V bench supply	owned
DRV1–DRV5	5	BIGTREETECH TMC2209 V1.3 + heatsink	
M1, M3	2	STEPPERONLINE 17HS19-1684S-PG5	J1, J3
M2	1	STEPPERONLINE 17HS19-1684S-PG14	J2
M4	1	STEPPERONLINE 17HS08-1004S	J4
M5	1	STEPPERONLINE 11HS20-0674S	J5
W7	1	MG996R servo	J6 gripper tool
J6	1	GX12 6-pin tool plug	
U6	1	XL4015 5 A buck, set 6.0 V	
ENC1–ENC5	5	AS5600 magnetic encoder module + diametric magnet	
F0–F2	3	Inline ATO fuse holders, 10 A, 2 A and 3 A fuses	
SW1	1	E-stop, 22 mm latching mushroom, 1 NC	
K1	1	24 V coil relay, normally open, with socket	
D1	1	1N4007 flyback diode	
C7, C20–C24	6	Electrolytic 1000 uF 16 V (x1), 100 uF 35 V (x5)	
C1–C5, C8	6	Ceramic 100 nF	
R1–R5, R13, R21–R25, R31–R32	13	1k (x8), 10k (x5) metal film 1/4 W	
FAN1	1	40 mm 5 V fan over driver row	

NOTES
1. Mega pins: STEP D22–26, DIR D30–34, EN D36–40 (LOW = driver on; 10k pull-ups keep drivers OFF while the Mega boots). encoders A0–A4, gripper PWM D9, tool I2C D20–D21.
2. TMC2209 standalone mode: UART mode: MS1/MS2 set each driver bus address (0–2). microstep is set over UART; CLK = GND selects the internal clock.
3. Check pin names against the silkscreen of the TMC2209 modules you buy (vendors differ). Set phase current with the module trimmer before connecting the arm.
4. TMC2209 VM max is about 29 V; 24 V only. Never plug or unplug a motor with 24 V present. Verify coil pairs with an ohmmeter.
5. Modules sit in female headers on perfboard. Heatsink on every module. 40 mm fan over the row.
6. All grounds common. Never connect 24 V or 6 V to the Mega 5 V pins.

Module wiring — not a validated machine
RoboArm
Sheet: /
File: custom_arm.kicad_sch
Title: RoboArm — full wiring
Size: A1 | Date: 2026-11-02 | Rev: D
KiCad E.D.A. 9.0.9 | Id: 1/1